RATIOS, RATES & PROPORTIONS

Full Solutions & Explanations

DOCUMENT 2 — SOLUTIONS & EXPLANATIONS

Total Questions: 40

Q1. Correct Answer: B • EASY

STEP-BY-STEP SOLUTION:

To simplify a ratio, divide both parts by their greatest common factor.
The greatest common factor of 8 and 12 is 4.
$8 \div 4 = 2$ and $12 \div 4 = 3$, so the simplified ratio is 2 : 3.

WHY EACH OPTION IS WRONG:

A) This divides both parts by 2 instead of by the greatest common factor (4), so it's equivalent in value but not yet in simplest form.
B) Correct. Dividing both 8 and 12 by their GCF of 4 gives 2 : 3.
C) This flips the order of the ratio, reversing flour and sugar.
D) This is the original ratio, not yet simplified at all.

TRAP ALERT 🪤
Stopping at a partially reduced ratio (like 4:6) instead of dividing by the GREATEST common factor, which is required to reach true simplest form.

KEY INSIGHT 💡
To fully simplify a ratio, always divide both terms by their greatest common factor, not just any common factor.

FAST METHOD ⚡
Find the GCF of both numbers first (here, GCF(8,12)=4), then divide both terms by it in one step.

Q2. Correct Answer: C • EASY

STEP-BY-STEP SOLUTION:

Cross-multiply to solve the proportion: $3 \times 20 = 4 \times x$.
$60 = 4x$.
$x = 60/4 = 15$.

WHY EACH OPTION IS WRONG:

A) This comes from an arithmetic slip while cross-multiplying or dividing.
B) This comes from incorrectly using a different fraction (such as 3/5) instead of the given 3/4.
C) Correct. Cross-multiplying gives $3(20) = 4x$, so $x = 60/4 = 15$.
D) This inverts the proportion, computing $20 \times 4/3$ instead of $20 \times 3/4$.

TRAP ALERT 🪤
Setting up the cross-multiplication backwards, which inverts the final answer.

KEY INSIGHT 💡
To solve $a/b = x/d$, cross-multiply to get $a \times d = b \times x$, then divide by b to isolate x.

FAST METHOD ⚡

Notice that 20 is 5 times 4 (the denominator on the left), so x must also be 5 times the numerator on the left: 3 × 5 = 15, with no cross-multiplication needed.

Q3. Correct Answer: D • EASY

STEP-BY-STEP SOLUTION:

Speed is a rate found by dividing distance by time: speed = distance/time.
speed = 240 miles / 4 hours.
speed = 60 miles per hour.

WHY EACH OPTION IS WRONG:

- **A)** This is the reciprocal of the correct rate (time/distance instead of distance/time).
- **B)** This simply repeats the given time value instead of computing the actual rate.
- **C)** This simply repeats the given distance value, skipping the division entirely.
- **D)** Correct. 240 miles ÷ 4 hours = 60 miles per hour.

TRAP ALERT 🪤

Reporting one of the given numbers directly instead of completing the division required to find a rate.

KEY INSIGHT 💡

A rate like speed always comes from dividing one quantity by another (here, distance by time) — never just restating one of the given numbers.

FAST METHOD ⚡

Divide directly: 240 ÷ 4 = 60 mph.

Q4. Correct Answer: A • EASY

STEP-BY-STEP SOLUTION:

Unit price is found by dividing the total cost by the total quantity: price per pound = total cost/number of pounds.
price per pound = $15/3 pounds.
price per pound = $5.

WHY EACH OPTION IS WRONG:

- **A)** Correct. $15 ÷ 3 pounds = $5 per pound.
- **B)** This is the reciprocal of the correct rate (pounds per dollar instead of dollars per pound).
- **C)** This adds the two given numbers (15 + 3 = 18) instead of dividing them.
- **D)** This multiplies the two given numbers (15 × 3 = 45) instead of dividing them.

TRAP ALERT 🪤

Combining the two given numbers with the wrong operation (adding or multiplying) instead of dividing, which is what a per-unit rate always requires.

KEY INSIGHT 💡

Any "per unit" rate (price per pound, miles per hour, etc.) is found by dividing the total amount by the number of units.

FAST METHOD ⚡

Divide total cost by quantity directly: \$15 ÷ 3 = \$5 per pound.

Q5. Correct Answer: 35 • EASY ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Cross-multiply: $5 \times 56 = 8 \times x$.
$280 = 8x$.
$x = 280/8 = 35$.

COMMON WRONG ANSWERS & WHY:

- Gridding 8.96 — dividing 56 by 5 instead of cross-multiplying correctly, then misapplying that result.
- Gridding 89.6 — multiplying 56 by 8/5 instead of by 5/8, inverting the proportion.

TRAP ALERT 🪤

Setting up the cross-multiplication with the numbers in the wrong positions, which inverts or scrambles the final result.

KEY INSIGHT 💡

Cross-multiplying $a/b = x/d$ always gives $a \times d = b \times x$ — keep careful track of which original numerator and denominator pair together.

FAST METHOD ⚡

Notice $56 \div 8 = 7$, so x must be 7 times the numerator on the left: $5 \times 7 = 35$, skipping the cross-multiplication.

Q6. Correct Answer: B • EASY

STEP-BY-STEP SOLUTION:

Set up a proportion: 1 inch/5 feet = 7 inches/x feet.
Cross-multiply: $1 \times x = 5 \times 7$.
$x = 35$ feet.

WHY EACH OPTION IS WRONG:

A) This adds the scale factor and the drawing distance ($5 + 7 = 12$) instead of multiplying them.
B) Correct. Since each inch represents 5 feet, 7 inches represents $7 \times 5 = 35$ feet.
C) This divides 7 by 5 instead of multiplying, inverting the scale relationship.
D) This divides 5 by 7 instead of multiplying, also inverting the scale relationship.

TRAP ALERT 🪤

Dividing instead of multiplying when scaling up from a drawing to real life — since each drawing unit represents MORE actual distance, the operation must be multiplication, not division.

KEY INSIGHT 💡

Scale drawings convert with a simple multiplication: (drawing measurement) × (actual distance per unit) = actual measurement.

FAST METHOD ⚡

Multiply the drawing distance directly by the scale factor: $7 \times 5 = 35$.

Q7. Correct Answer: C • EASY

STEP-BY-STEP SOLUTION:

The ratio 3:5 means the marbles are divided into 3+5 = 8 equal parts total.
Each part represents 40/8 = 5 marbles.
Red marbles make up 3 of those parts: 3 × 5 = 15 marbles.

WHY EACH OPTION IS WRONG:

A) This uses 3/5 of the total (a common but incorrect setup) instead of 3/8 of the total.
B) This computes some other incorrect fraction of the total, not matching the actual ratio parts.
C) Correct. With 8 total parts and 40 marbles, each part is 5; red marbles = 3 × 5 = 15.
D) This simply reports the "3" from the ratio directly, without scaling it to match the actual total of 40 marbles.

TRAP ALERT 🪤

Using the ratio number (3) directly as a fraction of the total over the OTHER ratio number (5) instead of over the SUM of both ratio numbers (8).

KEY INSIGHT 💡

When a ratio describes parts of a whole, always add the ratio terms together first to find the total number of equal parts, then scale to the actual total.

FAST METHOD ⚡

Add the ratio terms (3+5=8), divide the total by that sum to find one part's value (40/8=5), then multiply by the relevant ratio term (3×5=15).

Q8. Correct Answer: D • EASY

STEP-BY-STEP SOLUTION:

Rate is found by dividing total output by total time: rate = 150 widgets/6 hours.
rate = 25 widgets per hour.

WHY EACH OPTION IS WRONG:

A) This multiplies the two given numbers (150 × 6 = 900) instead of dividing them.
B) This subtracts the two given numbers (150 − 6 = 144), which has no meaningful connection to a rate.
C) This simply repeats the given number of hours instead of computing the actual rate.
D) Correct. 150 ÷ 6 = 25 widgets per hour.

TRAP ALERT 🪤

Combining the two given numbers with the wrong operation instead of recognizing that "per hour" always signals division.

KEY INSIGHT 💡

Any rate described as "per [unit of time]" is found by dividing the total amount by the total time.

FAST METHOD ⚡

Divide directly: 150 ÷ 6 = 25.

Q9. Correct Answer: 15 • EASY ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Set up a proportion: 6 eggs/4 people = x eggs/10 people.
Cross-multiply: $6 \times 10 = 4 \times x$.
$60 = 4x$.
$x = 60/4 = 15$.

COMMON WRONG ANSWERS & WHY:

• Gridding 6.67 — dividing 10 by 6 incorrectly instead of setting up and solving the proportion properly.
• Gridding 24 — multiplying 6 by 4 instead of correctly cross-multiplying with the new number of people (10).

TRAP ALERT 🪤

With three numbers given (4, 6, and 10), it's easy to combine the wrong pair or use the wrong operation when setting up the proportion.

KEY INSIGHT 💡

Always set up a proportion with matching units in matching positions (eggs over people on both sides), then cross-multiply to solve.

FAST METHOD ⚡

Find the per-person rate first (6 eggs ÷ 4 people = 1.5 eggs per person), then multiply by the new total: $1.5 \times 10 = 15$.

Q10. Correct Answer: A • EASY

STEP-BY-STEP SOLUTION:

The ratio 5:7 means the class is divided into 5+7 = 12 equal parts total.
Each part represents 36/12 = 3 students.
Girls make up 7 of those parts: $7 \times 3 = 21$ students.

WHY EACH OPTION IS WRONG:

A) Correct. With 12 total parts and 36 students, each part is 3; girls = $7 \times 3 = 21$.
B) This computes the number of BOYS ($5 \times 3 = 15$) instead of girls, mixing up which ratio term applies.
C) This simply repeats the total number of students instead of finding the girls-only portion.
D) This uses an incorrect fraction (7/10 instead of 7/12) of the total.

TRAP ALERT 🪤

Computing the correct method but applying it to the wrong ratio term — finding boys instead of the girls the question actually asked for.

KEY INSIGHT 💡

After finding the value of one equal part, always double-check which ratio term corresponds to what the question is actually asking for.

FAST METHOD ⚡

Add the ratio terms (5+7=12), divide the total by that sum (36/12=3), then multiply by the term for girls (7×3=21).

Q11. Correct Answer: B • MEDIUM

STEP-BY-STEP SOLUTION:

To combine the ratios, the value representing B must match in both ratios. Currently B is 3 in the first ratio and 4 in the second.

The least common multiple of 3 and 4 is 12. Scale the first ratio (2:3) by 4 to get 8:12, and scale the second ratio (4:5) by 3 to get 12:15.

Now both ratios share the same B value (12), so they combine into A:B:C = 8:12:15.

The ratio of A to C is 8:15.

WHY EACH OPTION IS WRONG:

A) This is the unsimplified intermediate ratio for A:B (8:12 reduces to 2:3) rather than the requested A:C ratio.

B) Correct. Scaling both ratios so B matches (12) gives A:B:C = 8:12:15, so A:C = 8:15.

C) This naively combines only the first and last numbers of the two original ratios (2 and 5) without first scaling B to a common value.

D) This results from a different incomplete combination that doesn't properly match the B terms first.

TRAP ALERT 🪤

Trying to combine two ratios by just grabbing the first and last numbers directly, without first scaling both ratios so that the shared term (B) matches in both.

KEY INSIGHT 💡

To combine two ratios that share a common term, scale each ratio so that the shared term has the same value in both, THEN read off the combined three-part ratio.

FAST METHOD ⚡

Find the LCM of the two B-values, scale each ratio up to that LCM, and the combined ratio falls right out.

Q12. Correct Answer: C ● MEDIUM

STEP-BY-STEP SOLUTION:

Using the scale, the actual distance in centimeters is 3.4 × 50,000 = 170,000 centimeters.

Convert centimeters to kilometers: there are 100,000 centimeters in 1 kilometer (100 cm/m × 1,000 m/km).

170,000 cm ÷ 100,000 = 1.7 kilometers.

WHY EACH OPTION IS WRONG:

A) This stops after finding the distance in centimeters, without converting to kilometers at all.

B) This divides by the wrong conversion factor (10,000 instead of 100,000), shifting the decimal point incorrectly.

C) Correct. 3.4 × 50,000 = 170,000 cm, and 170,000 ÷ 100,000 = 1.7 km.

D) This divides by an even larger incorrect factor, shifting the decimal point too far in the other direction.

TRAP ALERT 🪤

Forgetting the unit conversion step entirely after applying the scale factor, or using an incorrect number of centimeters per kilometer.

KEY INSIGHT 💡

Map scale problems often require TWO steps: first apply the scale factor to get a real-world distance in the original unit, then convert to whatever unit the question actually asks for.

FAST METHOD ⚡

Multiply by the scale factor first (3.4 × 50,000 = 170,000 cm), then move the decimal point 5 places left to convert to km (since 100,000 = 10^5).

Q13. Correct Answer: 4 • MEDIUM ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Express each pipe's rate as a fraction of the tank filled per hour: pipe A fills 1/6 of the tank per hour, and pipe B fills 1/12 per hour.

Working together, their combined rate is the sum: 1/6 + 1/12 = 2/12 + 1/12 = 3/12 = 1/4 of the tank per hour.

If 1/4 of the tank is filled each hour, the full tank takes 4 hours.

COMMON WRONG ANSWERS & WHY:

• Gridding 9 — simply averaging the two individual times ((6+12)/2 = 9), which is not how combined work rates behave.

• Gridding 18 — adding the two individual times directly (6+12=18) instead of combining their rates.

TRAP ALERT 🪤

Averaging or adding the TIMES directly instead of converting each to a RATE first, combining the rates, and then converting back to time at the very end.

KEY INSIGHT 💡

Combined work-rate problems always go through rates (work per unit time), never raw times directly: combined rate = $\text{rate}_1 + \text{rate}_2$, and combined time = 1/(combined rate).

FAST METHOD ⚡

Combined time for two workers/pipes equals $(\text{time}_1 \times \text{time}_2)/(\text{time}_1 + \text{time}_2)$: (6×12)/(6+12) = 72/18 = 4.

Q14. Correct Answer: D • MEDIUM

STEP-BY-STEP SOLUTION:

Find the actual amount of salt from each solution: Solution A contributes 0.20 × 4 = 0.8 liters of salt. Solution B contributes 0.50 × 2 = 1.0 liter of salt.

Total salt = 0.8 + 1.0 = 1.8 liters. Total mixture volume = 4 + 2 = 6 liters.

Percent salt in the mixture = 1.8/6 = 0.30 = 30%.

WHY EACH OPTION IS WRONG:

A) This takes the simple arithmetic average of 20% and 50% ((20+50)/2 = 35%), ignoring that the two solutions are mixed in DIFFERENT amounts (4 liters vs. 2 liters), not equal amounts.

B) This results from a different incorrect averaging or weighting of the two percentages.

C) This adds the two percentages directly (20% + 50% = 70%), which has no valid meaning in a mixture context.

D) Correct. Total salt (1.8 L) divided by total volume (6 L) gives 1.8/6 = 0.30 = 30%.

TRAP ALERT 🪤

Taking the simple average of the two percentages instead of a WEIGHTED average that accounts for the different volumes of each solution being mixed.

KEY INSIGHT 💡

Mixture percent problems require finding the actual amount of the substance in each part (percent × volume), summing those amounts, and dividing by the total combined volume — never just averaging the percentages.

FAST METHOD ⚡

Since 4 liters (twice as much) of the weaker solution is used, the mixture should be closer to 20% than to 50% — 30% fits that expectation, while 35% (the simple average) would not account for the unequal amounts.

Q15. Correct Answer: A • MEDIUM

STEP-BY-STEP SOLUTION:

Convert miles per hour to feet per second using two conversion factors: miles to feet (multiply by 5,280) and hours to seconds (divide by 3,600).

45 miles/hour × 5,280 feet/mile ÷ 3,600 seconds/hour.

45 × 5,280 = 237,600. Then 237,600 ÷ 3,600 = 66 feet per second.

WHY EACH OPTION IS WRONG:

A) Correct. 45 × 5,280 ÷ 3,600 = 66 feet per second.

B) This simply repeats the original speed value without performing any unit conversion at all.

C) This results from using an incorrect or incomplete conversion factor.

D) This results from multiplying by an incorrect conversion factor instead of properly combining both conversions.

TRAP ALERT 🪤

Forgetting to apply BOTH conversion factors (miles-to-feet AND hours-to-seconds), or applying only one of them.

KEY INSIGHT 💡

Converting a rate between unit systems (like mph to ft/s) requires multiplying by every necessary conversion factor in sequence — skipping any one factor gives an answer that's off by that factor.

FAST METHOD ⚡

Memorize that 60 mph = 88 ft/s as a benchmark; since 45 is 3/4 of 60, the answer should be roughly 3/4 of 88 ≈ 66, confirming the result quickly.

Q16. Correct Answer: B • MEDIUM

STEP-BY-STEP SOLUTION:

Direct variation means $y = kx$ for some constant k.

Use the given values to find k: $20 = k(4)$, so $k = 5$.

Now find y when $x = 9$: $y = 5(9) = 45$.

WHY EACH OPTION IS WRONG:

A) This results from using an incorrect constant of variation.

B) Correct. $k = 20/4 = 5$, so when $x = 9$, $y = 5(9) = 45$.

C) This simply repeats the original y value without applying the variation relationship to the new x value.

D) This divides instead of multiplying by the constant of variation, inverting the relationship.

TRAP ALERT 🪤

Finding the constant of variation correctly but then dividing by it instead of multiplying when computing the new y value, or skipping the constant-finding step entirely.

KEY INSIGHT 💡

Direct variation always follows $y = kx$, where k stays constant: first solve for k using the given pair of values, then use that same k for any other input.

FAST METHOD ⚡

Since x increased from 4 to 9 (a different, non-whole-number scale factor here), it's cleaner to find $k = 5$ first, then multiply: $5 \times 9 = 45$.

Q17. Correct Answer: **6.25** • MEDIUM ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Set up a proportion: 2.5 cups/6 servings = x cups/15 servings.
Cross-multiply: $2.5 \times 15 = 6 \times x$.
$37.5 = 6x$.
$x = 37.5/6 = 6.25$ cups.

COMMON WRONG ANSWERS & WHY:

- Gridding 6 — rounding the answer down to a whole number instead of keeping the precise decimal value.
- Gridding 5.6 — an arithmetic slip while cross-multiplying or dividing, possibly dividing 15 by 2.5 incorrectly with 6 misplaced.

TRAP ALERT 🪤

With a non-whole-number scale factor (15 servings is 2.5 times 6 servings), it's tempting to round the final answer to a cleaner-looking whole number instead of reporting the precise decimal value the proportion actually gives.

KEY INSIGHT 💡

Scaling proportions doesn't require whole-number results — keep all decimal precision through to the final answer unless the question specifically asks for rounding.

FAST METHOD ⚡

Find the scale factor first: $15 \div 6 = 2.5$, then multiply the original rice amount by that same factor: $2.5 \times 2.5 = 6.25$.

Q18. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

Population density is found by dividing total population by total area: density = population/area.
density = 18,000 people/12 square miles.
density = 1,500 people per square mile.

WHY EACH OPTION IS WRONG:

A) This is the reciprocal of the correct rate (area per person instead of people per area).
B) This multiplies the two given numbers ($18{,}000 \times 12 = 216{,}000$) instead of dividing them.
C) Correct. $18{,}000 \div 12 = 1{,}500$ people per square mile.

D) This is off by a factor of 10, likely from a division error involving a misplaced decimal point.

TRAP ALERT 🪤

Multiplying the two given quantities instead of dividing, or making a decimal-placement error during division.

KEY INSIGHT 💡

Density (population per area, mass per volume, etc.) always comes from dividing the total amount by the total space it occupies.

FAST METHOD ⚡

Divide directly: 18,000 ÷ 12 = 1,500.

Q19. Correct Answer: D • MEDIUM

STEP-BY-STEP SOLUTION:

Find each runner's average speed by dividing distance by time.
Runner A: 100 meters/12 seconds ≈ 8.33 meters per second.
Runner B: 150 meters/17 seconds ≈ 8.82 meters per second.
Since 8.82 > 8.33, Runner B has the greater average speed and is faster.

WHY EACH OPTION IS WRONG:

A) Comparing the raw time values alone (without dividing by distance) doesn't account for the fact that the two runners covered different distances, so this comparison is meaningless on its own.
B) The two speeds are close but not exactly equal; calculating each precisely shows Runner B is faster.
C) Comparing the raw distance values alone (without dividing by time) is just as meaningless as comparing times alone.
D) Correct. Runner B's speed (≈ 8.82 m/s) exceeds Runner A's speed (≈ 8.33 m/s).

TRAP ALERT 🪤

Comparing only one of the two given numbers (just the times, or just the distances) instead of computing the actual rate (distance ÷ time) for each runner before comparing.

KEY INSIGHT 💡

Whenever two scenarios involve different amounts of both quantities (here, both distance AND time differ), you must compute the actual RATE for each before any valid comparison can be made.

FAST METHOD ⚡

Convert each to a per-second rate and compare directly — with a calculator, this takes only two quick divisions.

Q20. Correct Answer: A • MEDIUM

STEP-BY-STEP SOLUTION:

The ratio 1:12 means there is 1 manager for every 12 employees.
Set up a proportion: 1 manager/12 employees = x managers/156 employees.
Cross-multiply: 1 × 156 = 12 × x, so 156 = 12x.
x = 156/12 = 13 managers.

WHY EACH OPTION IS WRONG:

A) Correct. 156 ÷ 12 = 13 managers.
B) This simply repeats the "12" from the given ratio instead of completing the proportion calculation.
C) This simply repeats the total number of employees instead of finding the proportional number of managers.
D) This multiplies the two given numbers (156 × 12) instead of dividing them.

TRAP ALERT 🪤

Multiplying instead of dividing, or stopping partway through the calculation and reporting a number directly from the problem instead of the actual computed answer.

KEY INSIGHT 💡

When a ratio compares a small group to a larger one (like 1 manager per 12 employees), divide the larger total by the ratio's second term to find the matching count for the first term.

FAST METHOD ⚡

Divide directly: 156 ÷ 12 = 13.

Q21. Correct Answer: B • MEDIUM

STEP-BY-STEP SOLUTION:

First find the rate per page: \$32.50/250 pages = \$0.13 per page.
Multiply this rate by the new number of pages: \$0.13 × 880 pages.
Total cost = \$114.40.

WHY EACH OPTION IS WRONG:

A) This results from using a slightly incorrect per-page rate before multiplying.
B) Correct. Rate = \$32.50/250 = \$0.13 per page; cost for 880 pages = \$0.13 × 880 = \$114.40.
C) This simply repeats the original given cost without scaling it to the new number of pages at all.
D) This sets up the proportion backwards, effectively dividing by the rate instead of multiplying by it.

TRAP ALERT 🪤

Finding the correct per-page rate but then either forgetting to apply it to the new quantity or applying it with the wrong operation (dividing instead of multiplying).

KEY INSIGHT 💡

Cost-scaling problems with a constant rate always follow the same two-step pattern: find the unit rate first (cost ÷ quantity), then multiply that rate by the new quantity.

FAST METHOD ⚡

Find the unit rate once (\$32.50 ÷ 250 = \$0.13), then multiply directly by the new page count: \$0.13 × 880 = \$114.40.

Q22. Correct Answer: 42 • MEDIUM ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Find the rate per worker: 18 chairs/3 workers = 6 chairs per worker per day.
Multiply this rate by the new number of workers: 6 chairs/worker × 7 workers.

Total = 42 chairs.

COMMON WRONG ANSWERS & WHY:

- Gridding 21 — incorrectly halving or otherwise miscalculating the per-worker rate before scaling up.
- Gridding 126 — multiplying 18 by 7 directly without first dividing by 3 to find the per-worker rate.

TRAP ALERT 🪤

Multiplying the given total (18 chairs) directly by the new number of workers (7) without first finding the per-worker rate, which leads to a result that's 3 times too large.

KEY INSIGHT 💡

Worker-rate problems require finding the individual rate first (total ÷ number of workers), then scaling that individual rate to the new group size.

FAST METHOD ⚡

Find the per-worker rate (18÷3=6), then multiply by the new worker count (6×7=42) in two clean steps.

Q23. Correct Answer: C • MEDIUM

STEP-BY-STEP SOLUTION:

Convert the total time to minutes: 1 hour 12 minutes = 60 + 12 = 72 minutes.
Multiply the rate by the total time: 2.5 gallons/minute × 72 minutes.
Total = 180 gallons.

WHY EACH OPTION IS WRONG:

A) This uses only the 60-minute portion of the time, forgetting to include the extra 12 minutes.
B) This results from a different incomplete time calculation.
C) Correct. 72 minutes × 2.5 gallons/minute = 180 gallons.
D) This divides instead of multiplying, inverting the rate relationship entirely.

TRAP ALERT 🪤

Forgetting to convert "1 hour 12 minutes" fully into a single unit (minutes) before multiplying, and instead only using part of the given time.

KEY INSIGHT 💡

Whenever a time is given in mixed units (like hours and minutes), convert it entirely into one consistent unit before applying any rate.

FAST METHOD ⚡

Convert hours to minutes first (1 hour = 60 minutes, plus the extra 12 = 72 total), then multiply straight through: 72 × 2.5 = 180.

Q24. Correct Answer: D • MEDIUM

STEP-BY-STEP SOLUTION:

Let the number of boys be 3k and the number of girls be 4k, for some positive value k.
The difference between girls and boys is 4k − 3k = k, and this difference is given as 15, so k = 15.
The number of boys is 3k = 3(15) = 45.

WHY EACH OPTION IS WRONG:

A) This reports the value of k itself (15) instead of using it to compute the actual number of boys (3k).

B) This results from a different incomplete calculation that doesn't correctly scale k by the boys' ratio term.

C) This computes the number of GIRLS (4k = 60) instead of the boys the question asked for.

D) Correct. k = 15 (since 4k−3k=k=15), so boys = 3k = 3(15) = 45.

TRAP ALERT 🪤

Solving correctly for k but then either reporting k itself as the final answer, or applying it to the wrong ratio term (girls instead of boys).

KEY INSIGHT 💡

When a ratio problem gives a DIFFERENCE between two groups (rather than a total), the difference between the ratio terms themselves equals the scale factor k — then apply k to whichever specific group the question asks about.

FAST METHOD ⚡

Subtract the ratio terms (4−3=1 part = the given difference), so k equals the difference directly here; then multiply by the boys' ratio term (3) to finish.

Q25. Correct Answer: A • MEDIUM

STEP-BY-STEP SOLUTION:

Pipe A's rate is 1/8 of the pool per hour. The combined rate of A and B together is 1/3 of the pool per hour.

Pipe B's rate alone is the combined rate minus pipe A's rate: 1/3 − 1/8.

Find a common denominator (24): 8/24 − 3/24 = 5/24.

Pipe B fills 5/24 of the pool per hour, so alone it takes 24/5 = 4.8 hours.

WHY EACH OPTION IS WRONG:

A) Correct. Pipe B's rate is 1/3 − 1/8 = 5/24 per hour, so its alone time is 24/5 = 4.8 hours.

B) This results from a rounding or estimation shortcut instead of the precise calculation.

C) This adds the two given times directly (8 + 3 = 11), which has no valid meaning for combined work rates.

D) This results from an inverted or otherwise incorrect setup of the rate subtraction.

TRAP ALERT 🪤

Trying to combine or subtract the given TIMES directly (8 hours and 3 hours) instead of first converting both to RATES, subtracting the rates, and only then converting back to a time.

KEY INSIGHT 💡

To find one worker's individual rate when you know the combined rate and the other worker's individual rate, subtract: rate(unknown) = rate(combined) − rate(known).

FAST METHOD ⚡

Once you have pipe B's rate (5/24), flip it to get the time directly: time = 1/rate = 24/5 = 4.8 hours.

Q26. Correct Answer: 490 • MEDIUM ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Find the train's speed: 315 miles/4.5 hours = 70 miles per hour.
Multiply this speed by the new time: 70 miles/hour × 7 hours.
Total distance = 490 miles.

COMMON WRONG ANSWERS & WHY:

• Gridding 45 — an arithmetic error while computing the original speed (315 ÷ 4.5).
• Gridding 202.5 — dividing 315 by 7 directly instead of first finding the rate and then multiplying by 7.

TRAP ALERT 🪤

With a decimal value in the original time (4.5 hours), it's easy to make an arithmetic slip while finding the unit rate before scaling to the new time.

KEY INSIGHT 💡

Constant-speed problems always follow the same pattern: find the rate first (distance ÷ time), then multiply that rate by whatever new time value is given.

FAST METHOD ⚡

315 ÷ 4.5 simplifies cleanly if you double both numbers first: 630/9 = 70 mph, then 70 × 7 = 490.

Q27. Correct Answer: B • MEDIUM

STEP-BY-STEP SOLUTION:

Find the unit price for each brand: Brand A: $3.20/16 ounces = $0.20 per ounce.
Brand B: $4.32/24 ounces = $0.18 per ounce.
Since $0.18 is less than $0.20, Brand B offers the lower price per ounce.

WHY EACH OPTION IS WRONG:

A) This correctly computes Brand A's unit price but incorrectly identifies it as the lower one, when Brand B's price is actually lower.

B) Correct. Brand A costs $0.20/oz and Brand B costs $0.18/oz, so Brand B is cheaper per ounce.

C) This correctly identifies Brand A but pairs it with Brand B's actual unit price, mismatching the brand and value.

D) This correctly identifies Brand B as cheaper but pairs it with Brand A's actual unit price, mismatching the brand and value.

TRAP ALERT 🪤

Computing both unit prices correctly but then mismatching which price belongs to which brand when selecting a final answer.

KEY INSIGHT 💡

"Best buy" comparison problems require computing a true unit rate (price per ounce, per pound, etc.) for EACH option before any valid comparison can be made — total price alone is misleading when package sizes differ.

FAST METHOD ⚡

Divide each price by its respective quantity, then simply compare the two resulting unit prices directly: smaller unit price wins.

Q28. Correct Answer: C • MEDIUM

STEP-BY-STEP SOLUTION:

First find the amount of water in the bottle: 750 − 250 = 500 milliliters of water.
The ratio of pure juice to water is 250:500.
Divide both terms by their greatest common factor, 250: 250/250 : 500/250 = 1:2.

WHY EACH OPTION IS WRONG:

A) This compares pure juice to the TOTAL bottle volume (250:750 = 1:3) instead of to the water amount specifically.

B) This incorrectly assumes the bottle splits evenly between juice and water without actually computing the water amount.

C) Correct. Water = 750−250 = 500 mL, so juice:water = 250:500 = 1:2.

D) This flips the correct ratio, reversing the order of juice and water.

TRAP ALERT 🪤

Forgetting to first subtract to find the water amount, and instead comparing the juice amount directly to the TOTAL bottle volume, which represents a different ratio entirely (juice to total, not juice to water).

KEY INSIGHT 💡

When a ratio problem describes a "part" and a "whole," carefully identify whether you need part-to-whole or part-to-OTHER-part — these require different setups.

FAST METHOD ⚡

Subtract first to isolate both individual amounts (juice and water separately), then form the ratio directly between those two numbers — never the part and the total together when a part-to-part ratio is requested.

Q29. Correct Answer: D • HARD

STEP-BY-STEP SOLUTION:

Since the "oranges" value already matches in both ratios (4 in each), they can be combined directly: apples : oranges : pears = 3 : 4 : 5.
The total number of equal parts is 3 + 4 + 5 = 12.
Each part represents 180/12 = 15 pieces of fruit.
Pears make up 5 of those parts: 5 × 15 = 75 pears.

WHY EACH OPTION IS WRONG:

A) This results from using an incorrect ratio term or part value for pears.

B) This results from an arithmetic slip while computing the per-part value or applying it to pears.

C) This computes the number of APPLES (3 × 15 = 45) instead of pears, mixing up the ratio terms.

D) Correct. Combined ratio 3:4:5 has 12 total parts; 180/12=15 per part; pears = 5×15 = 75.

TRAP ALERT 🪤

Forgetting to double-check that the shared term (oranges) already matches between the two ratios before combining — here it conveniently already does (4 in both), but missing this check could lead to incorrectly combining ratios that need scaling first.

KEY INSIGHT 💡

When combining two ratios that share a common term, always verify whether that shared term already matches numerically; if it does, the three-part ratio combines directly with no extra scaling needed.

FAST METHOD ⚡

Since the middle term (oranges = 4) already matches in both given ratios, just write the three-part ratio straight across: 3:4:5, then divide the total by the part-sum (12) and multiply by the relevant term.

Q30. Correct Answer: A • HARD

STEP-BY-STEP SOLUTION:

Maria's rate is 1/8 of the fence per hour, and Jamal's rate is 1/12 per hour. Their combined rate is 1/8 + 1/12.

Using a common denominator of 24: 3/24 + 2/24 = 5/24 of the fence per hour, working together.

In 3 hours together, they complete 3 × 5/24 = 15/24 = 5/8 of the fence.

The remaining portion is 1 − 5/8 = 3/8 of the fence.

Jamal alone works at 1/12 per hour, so the time needed is (3/8) ÷ (1/12) = (3/8) × 12 = 36/8 = 4.5 hours.

WHY EACH OPTION IS WRONG:

A) Correct. After 3 hours together, 5/8 of the fence is done, leaving 3/8; at Jamal's rate of 1/12 per hour, that takes 4.5 hours.

B) This results from a rounding or estimation shortcut instead of the precise fraction calculation.

C) This ignores the work already completed during the 3 hours together, and instead uses Jamal's FULL original time (close to 12, adjusted incorrectly) as if starting from scratch.

D) This results from an inverted setup of the final division step.

TRAP ALERT 🪤

Forgetting to first subtract the portion of work already completed during the 3 hours of joint effort, and instead either using Jamal's full original 12-hour time or miscalculating the remaining fraction.

KEY INSIGHT 💡

Multi-stage work problems require tracking exactly how much of the job is complete at each stage before calculating how much time remains — always work in fractions of the whole job, not raw hours, until the very last step.

FAST METHOD ⚡

Find the combined rate, multiply by the joint working time to get the completed fraction, subtract from 1 for the remaining fraction, then divide that remaining fraction by the SOLO worker's individual rate.

Q31. Correct Answer: 5 • HARD ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Let x represent the number of liters of the 80% solution needed.

Set up an equation based on the total amount of pure acid: 0.80x + 0.20(5) = 0.50(x + 5).

Simplify: 0.80x + 1 = 0.50x + 2.5.

Subtract 0.50x from both sides and subtract 1 from both sides: 0.30x = 1.5.

Divide: x = 1.5/0.30 = 5 liters.

COMMON WRONG ANSWERS & WHY:

• Gridding 2.5 — averaging the two given volumes or percentages incorrectly instead of solving the full mixture equation.

• Gridding 10 — setting up the equation with the percentages or volumes swapped, doubling the correct result.

TRAP ALERT 🪤

Trying to find the answer through estimation or simple averaging instead of setting up and solving the full mixture equation, which is the only reliable method when the target percentage isn't exactly halfway between the two given percentages relative to volume.

KEY INSIGHT 💡

Mixture problems always follow the pattern: (amount of pure substance from solution 1) + (amount of pure substance from solution 2) = (amount of pure substance in the final mixture), where each "amount of pure substance" is percentage × volume.

FAST METHOD ⚡

Set up the equation with one unknown (the volume you need to find), expand, and solve like any other linear equation — resist the urge to guess based on how close the percentages look to each other.

Q32. Correct Answer: B • HARD

STEP-BY-STEP SOLUTION:

Average speed is total distance divided by total time — NOT the simple average of the two speeds.

Let the one-way distance be d. The time going is d/12, and the time returning is d/18.

Total distance for the round trip is 2d. Total time is d/12 + d/18, which combines (using a common denominator of 36) to 3d/36 + 2d/36 = 5d/36.

Average speed = total distance/total time = 2d/(5d/36) = 2d × 36/(5d) = 72/5 = 14.4 mph.

WHY EACH OPTION IS WRONG:

A) This takes the simple arithmetic average of the two speeds ((12+18)/2 = 15), which is INCORRECT whenever the time spent at each speed differs — here, more time is spent at the slower speed, pulling the true average down.

B) Correct. Using total distance ÷ total time gives 72/5 = 14.4 mph.

C) This results from a different incorrect averaging or estimation method.

D) This results from yet another incorrect averaging method, overshooting in the other direction.

TRAP ALERT 🪤

Averaging the two given speeds directly — this is the single most common trap in average-speed problems and is ALWAYS wrong unless equal amounts of TIME (not distance) are spent at each speed.

KEY INSIGHT 💡

True average speed always equals total distance divided by total time. When equal DISTANCES are traveled at two different speeds (as in a round trip), the average speed is the harmonic mean of the two speeds, which is always less than the simple arithmetic average.

FAST METHOD ⚡

For a round trip with equal distances at speeds v_1 and v_2, average speed = $2(v_1)(v_2)/(v_1+v_2)$: 2(12)(18)/(12+18) = 432/30 = 14.4.

Q33. Correct Answer: C • HARD

STEP-BY-STEP SOLUTION:

During the first 4 hours, both machines run together, producing a combined (45 + 60) = 105 parts per hour.
Parts produced in the first 4 hours: 4 × 105 = 420 parts.
During the next 2 hours, only Machine Y runs, producing 60 parts per hour.
Parts produced in the next 2 hours: 2 × 60 = 120 parts.
Total parts = 420 + 120 = 540.

WHY EACH OPTION IS WRONG:

A) This accounts only for the first 4-hour period and forgets to add the additional 2 hours of Machine Y running alone.

B) This results from an incomplete or miscalculated addition of the two time periods.

C) Correct. 4(45+60) + 2(60) = 420 + 120 = 540.

D) This results from incorrectly running both machines for the full 6 hours, ignoring that Machine X actually broke down after 4 hours.

TRAP ALERT 🪤

Forgetting that the production rate CHANGES partway through the problem — either ignoring the breakdown entirely (treating both machines as running the full 6 hours) or forgetting to add the final 2-hour single-machine period at all.

KEY INSIGHT 💡

Multi-stage rate problems must be broken into separate time periods whenever the rate changes, with each period's production calculated separately and then summed at the end.

FAST METHOD ⚡

Calculate each distinct time period separately using its own specific rate, then add all the periods together as the very last step — never apply one single rate across the entire combined time.

Q34. Correct Answer: D • HARD

STEP-BY-STEP SOLUTION:

Let x = 5k and y = 7k for some positive value k, based on the original ratio.
After both values increase by 10, the new ratio is (5k+10)/(7k+10) = 3/4.
Cross-multiply: 4(5k+10) = 3(7k+10).
Expand: 20k + 40 = 21k + 30.
Subtract 20k from both sides and subtract 30 from both sides: 10 = k.
So x = 5k = 5(10) = 50.

WHY EACH OPTION IS WRONG:

A) This results from an arithmetic slip while solving the linear equation for k.

B) This reports the value of k itself (10) instead of using it to compute x = 5k.

C) This results from a different incomplete or incorrect calculation.

D) Correct. Solving 4(5k+10)=3(7k+10) gives k=10, so x = 5k = 50.

TRAP ALERT 🪤

Solving correctly for k but then reporting k itself as the final answer, forgetting that the question asks for x (which equals 5k), not k directly.

KEY INSIGHT 💡

When a ratio problem introduces a variable like k to represent the original ratio terms, always re-read the question carefully at the end to determine whether it's asking for k itself or for one of the original quantities (which requires one more multiplication step).

FAST METHOD ⚡

Set up x=5k and y=7k from the start, build the equation directly from the new ratio condition, solve for k, and only then compute the SPECIFIC value the question asks for (here, x, not k).

Q35. Correct Answer: 3 ● HARD ✎ GRID-IN

STEP-BY-STEP SOLUTION:

The total amount of work, measured in "worker-days," stays constant regardless of how many workers are used: 5 workers × 9 days = 45 worker-days of total work.
With 15 workers sharing this same total workload, divide: 45 worker-days ÷ 15 workers = 3 days.

COMMON WRONG ANSWERS & WHY:

• Gridding 27 — multiplying 9 by 3 (the factor by which the workforce increased) instead of dividing, treating this as direct variation when it's actually inverse variation.
• Gridding 15 — simply gridding the new number of workers instead of computing the actual number of days required.

TRAP ALERT 🪤

Treating this as a direct variation problem (more workers means more days, so multiply) when it's actually an INVERSE variation problem (more workers means FEWER days, so divide), since the total work stays fixed.

KEY INSIGHT 💡

When more workers tackle the SAME fixed amount of work, the number of days decreases proportionally — this is inverse variation, captured by the constant product: (workers) × (days) = constant total work.

FAST METHOD ⚡

Recognize that the workforce tripled (5→15), so the time required must be exactly one-third of the original: 9 ÷ 3 = 3 days, with no need to compute total worker-days explicitly.

Q36. Correct Answer: A ● HARD

STEP-BY-STEP SOLUTION:

Inverse proportion means xy = k for some constant k.
Use the given values to find k: (15)(8) = 120, so k = 120.
Now find y when x = 24: 24y = 120.
y = 120/24 = 5.

WHY EACH OPTION IS WRONG:

A) Correct. k = 15×8 = 120, so when x=24, y = 120/24 = 5.
B) This incorrectly applies a DIRECT variation setup (y = kx with k from the original ratio) instead of the inverse relationship xy = k.
C) This results from a different incorrect setup, possibly involving a flipped or miscalculated ratio.
D) This results from computing the reciprocal relationship incorrectly, landing on an inflated value.

TRAP ALERT 🪤

Treating an inverse variation problem like a direct variation problem — in direct variation, y/x stays constant, but in inverse variation, it's the PRODUCT xy that stays constant instead.

KEY INSIGHT 💡

Inverse proportion: $xy = k$ (constant product). Direct proportion: $y/x = k$ (constant ratio). These are fundamentally different relationships, and using the wrong one gives a completely different (and incorrect) result.

FAST METHOD ⚡

Find the constant product first ($15 \times 8 = 120$), then divide that same constant by the new x value ($120 \div 24 = 5$) — always division for inverse variation, never multiplication by a ratio.

Q37. Correct Answer: B • HARD

STEP-BY-STEP SOLUTION:

First convert the actual car's length to inches: 16.2 feet × 12 inches/foot = 194.4 inches.
Apply the scale factor: model length = 194.4 inches ÷ 18.
Model length = 10.8 inches.

WHY EACH OPTION IS WRONG:

A) This applies the scale factor (÷18) to the ORIGINAL feet measurement without first converting to inches, producing an answer still in the wrong unit.
B) Correct. 16.2 feet = 194.4 inches; 194.4 ÷ 18 = 10.8 inches.
C) This is off by a factor of 10 from the correct answer, likely from a decimal placement error.
D) This multiplies instead of dividing by the scale factor, producing a model far LARGER than the actual car, which doesn't make sense for a 1:18 scale-down.

TRAP ALERT 🪤

Performing the unit conversion (feet to inches) and the scale-factor division in the wrong order, or forgetting the unit conversion step entirely, which leaves the final answer in the wrong units even if the scale math itself is correct.

KEY INSIGHT 💡

Scale-model problems involving a unit change require care about WHEN to convert: it's often cleanest to convert to the FINAL desired unit first, then apply the scale factor, so the result is already in the right units.

FAST METHOD ⚡

Convert to inches first (16.2 × 12 = 194.4), then divide by the scale denominator (194.4 ÷ 18 = 10.8) — doing the conversion before the scaling avoids needing a second conversion step at the end.

Q38. Correct Answer: C • HARD

STEP-BY-STEP SOLUTION:

Pipe A's filling rate is 1/10 per hour, and pipe B's filling rate is 1/15 per hour. Pipe C's DRAINING rate is 1/12 per hour, which must be SUBTRACTED since it works against the other two.
Combined net rate = $1/10 + 1/15 - 1/12$.
Using a common denominator of 60: $6/60 + 4/60 - 5/60 = 5/60 = 1/12$ of the pool per hour.
Since the pool fills at a net rate of 1/12 per hour, the full pool takes 12 hours.

WHY EACH OPTION IS WRONG:

- **A)** This treats Pipe C as another FILLING pipe instead of a drain, adding all three rates together (1/10+1/15+1/12=1/4) instead of subtracting the drain's rate.
- **B)** This results from a different incorrect combination of the three rates.
- **C)** Correct. Net rate = 1/10 + 1/15 − 1/12 = 1/12 per hour, so the pool fills in 12 hours.
- **D)** This results from an inverted or otherwise mishandled final calculation.

TRAP ALERT 🪤

Treating the drain (Pipe C) as if it were a third filling pipe and adding its rate instead of subtracting it — a drain works AGAINST the filling pipes, reducing the net rate rather than increasing it.

KEY INSIGHT 💡

When a problem combines pipes that fill AND pipes that drain, filling rates add together, but draining rates must be subtracted, since they work in the opposite direction.

FAST METHOD ⚡

Identify each pipe's role first (filling = positive rate, draining = negative rate), combine with correct signs, then take the reciprocal of the net rate for the final time.

Q39. Correct Answer: 4860 • HARD ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Use the administrators-to-teachers ratio to find the number of teachers: since the ratio is 1:6, teachers = 45 × 6 = 270.

Use the teachers-to-students ratio to find the number of students: since the ratio is 1:18, students = 270 × 18 = 4,860.

COMMON WRONG ANSWERS & WHY:

• Gridding 810 — multiplying 45 by 18 directly, skipping the intermediate step of finding the number of teachers first.

• Gridding 270 — stopping after finding the number of teachers and forgetting to complete the final step to find students.

TRAP ALERT 🪤

With two separate ratios chained together (administrators → teachers → students), it's easy to skip the middle step (finding teachers) and try to relate administrators directly to students, or to stop too early after finding only the teacher count.

KEY INSIGHT 💡

Chained ratio problems must be solved one link at a time, in order — use the first ratio to find the middle quantity, then use that result (not the original given number) in the second ratio.

FAST METHOD ⚡

Work through the chain step by step: administrators × 6 = teachers, then teachers × 18 = students — multiply straight through in sequence: 45 × 6 × 18 = 4,860.

Q40. Correct Answer: D • HARD

STEP-BY-STEP SOLUTION:

First find the amount of sugar needed for the original 24-person recipe. The ratio 5:2 means there are 5+2=7 total parts, and 4.2 cups represents all 7 parts.

Each part equals 4.2/7 = 0.6 cups.
Sugar (2 parts) for the original recipe = 2 × 0.6 = 1.2 cups.
Find the scale factor for the new serving size: 60 people/24 people = 2.5.
Scaled sugar amount = 1.2 × 2.5 = 3 cups.

WHY EACH OPTION IS WRONG:

A) This correctly finds the ORIGINAL sugar amount (1.2 cups, for 24 people) but forgets to scale it up for the new 60-person batch.

B) This computes the FLOUR amount (5 parts × 0.6 = 3.0, scaled by 2.5 = 7.5) instead of the sugar amount the question asks for.

C) This applies an incorrect, smaller scale factor (1.25 instead of 2.5), halving the correct scaling adjustment.

D) Correct. Original sugar = 1.2 cups; scale factor = 60/24 = 2.5; scaled sugar = 1.2 × 2.5 = 3 cups.

TRAP ALERT 🪤

This problem stacks two separate skills — splitting a combined total by a ratio, AND scaling a recipe to a new serving size — making it easy to do one step correctly but forget the other entirely.

KEY INSIGHT 💡

Multi-stage ratio and scaling problems should be solved in clearly separated steps: first resolve the ratio split at the ORIGINAL size, then apply any scaling factor as a completely separate final step.

FAST METHOD ⚡

Find one ratio part's value (4.2/7=0.6), multiply by sugar's share (2×0.6=1.2) for the original batch, then multiply by the serving scale factor (60/24=2.5) last: 1.2×2.5=3.

TOPIC SUMMARY — Ratios, Rates & Proportions

Common Mistakes:

1. Averaging instead of properly weighting: simple-averaging two rates, percentages, or speeds (like (12+18)/2 for a round-trip average speed, or (20%+50%)/2 for a mixture) almost always gives the wrong answer unless the two quantities being combined are weighted exactly equally.
2. Confusing direct and inverse variation: in direct variation, y/x stays constant (more workers → more output in the same time); in inverse variation, the PRODUCT stays constant (more workers → less time for the SAME fixed job). Using the wrong relationship inverts the entire answer.
3. Stopping one step early: reporting an intermediate value (like a scale factor k, a per-part amount, or one stage of a multi-step word problem) instead of carrying it through to the specific final quantity the question actually asks for.

Must Remember:

1. To combine two ratios sharing a common term (A:B and B:C), scale both ratios so the shared term matches, then read off the combined three-part ratio.
2. Average speed (or any rate) for a trip with unequal times at each speed is ALWAYS total distance ÷ total time — never the simple average of the speeds themselves.
3. Combined work-rate problems convert times to rates (1/time), add or subtract those rates as appropriate (subtracting for drains or opposing rates), and convert back to time (1/combined rate) only at the very end.

Expected on DSAT:

Based on this project's frequency analysis of past Digital SAT papers, ratios, rates, and proportions appeared in 11 of 1,240 math questions analyzed (about 0.9%). That works out to roughly 0–1 question per test on average, appearing in about 1 of every 3 tests — when it does show up, it's frequently dressed in a real-world context (recipes, speeds, mixtures, scale models) rather than presented as bare numbers.

— END OF SOLUTIONS —